

Sahas Munamala

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Education:

University of Illinois – Urbana-Champaign

B.S. Computer Eng.
May 2023 | Champaign, IL

Skills

- Distributed Systems
- Low-Latency Networking
- Cloud Security
- AWS
- Linux Kernel Interfaces
- Mathematical Simulation
- Writing

Certifications/Publications

- Alternating Causality (in proceedings)
- AWS Certified Solutions Architect

Languages

Python • Rust • C/C++ •
SystemVerilog • Java •
JavaScript

Tools & Frameworks

Github • React • Docker •
DynamoDB • Kubernetes •
OpenCV • FPGAs •
Raspberry Pi • Pytorch •
Linux/Bash • MySQL

Interests

Writing/Systems Research
Social Infra & Technical
Communities
Baking
Muay Thai
Distance Running

Experience

Founding Engineer | DÆDÆLUS

July 2023 - December 2025

- **Developed** a low-latency protocol stack on Xilinx FPGA hardware, implementing AXI-stream based packet processing and acknowledgment, achieving 20ns round trip latency at 1G with SystemVerilog and CocoTB.
- **Engineered** an emulation rig for rapid prototyping of new protocols over point-to-point thunderbolt in Python/Rust, with 15µs round trip latency on Mac Minis.
- **Validated** a custom reversible communication protocol over a Thunderbolt Mesh with a custom network dashboard and sub-ms failure detection and rerouting.

Project Lead | Open Atomic Ethernet (OCP)

January 2025 - December 2025

- **Co-Founded** the Open Atomic Ethernet workstream, driving early architecture, organizing collaboration with contributors, and setting the direction for the next 50 years of Ethernet.
- **Authored** technical papers and specifications on reliable layer 2 communication, atomic link-layer protocols, and graph algorithms.

Software Engineer | Amazon Rufus

November 2023 - April 2025

- **Shipped** a global rule deployment system impacting 120,000TPS for AI conversational safety, improving rollout reliability and cutting deployment time by 95%.
- **Automated** rule reporting and retirement, eliminating manual workflows and saving an estimated \$3M+ annually in operational overhead.
- **Built** cross-region rule replication and internationalization support, improving availability, fault tolerance, deployed across three regions.

Software Development Intern | Amazon Alexa

Summer 2022

- **Built, tested, and deployed** an interface to interpret 7 different Alexa Shopping language model outputs for internal analysis.
- **Implemented** new API endpoints to an AWS Serverless backend to support new features to the tool such as model overrides and switching language packs.
- **Conducted** user research, implemented enhancements with beta testing to production. These new features are used by managers, data scientists, and developers across the organization.

Projects

SNES FPGA Emulator *November 2025*

Built an FPGA-based NES emulator in SystemVerilog, integrating CPU, memory, and graphics components to reproduce game behavior in hardware.

Distributed Key Value Simulator *March 2026*

An NS-3-based distributed key value database simulator for studying how network conditions, failures, and topology changes create tail latency and long tail operational effects during infrastructure rollouts.